

Namibian hake model update, 2026

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Executive summary

Summary of changes in assessment inputs

The accepted 2025 model configuration was extended through 2026 without changing the base-model assumptions. The update adds the 2026 catch, CPUE, survey biomass, and composition rows provided in `namhakdata26.dat`. The species-specific survey series in `CapeHakes.csv` was extended through 2026 for *Merluccius paradoxus* and *M. capensis*.

Automated checks confirmed continuous annual ADMB matrices from 1964 through 2026, the expected dimensions of all weight, catch, composition, CPUE, and survey blocks, agreement among all terminal-year controls, and valid 54321 and 12345 end-of-file sentinels.

Summary of results

The 2026 base model converged with an objective function of -44.977, a maximum gradient of 4e-05, and a positive-definite Hessian. Estimated terminal spawning biomass was 467 and terminal spawning biomass relative to BMSY was 0.497.

Four of six sensitivity configurations met the same convergence criteria as the base model. The dome-shaped selectivity case had parameters on their lower bounds and a singular Hessian. The model with steepness fixed at 0.5 had a maximum gradient above 0.001 after an extended optimization. Results from those two configurations are not used for inference.

No TAC, ABC, or OFL recommendation is made in this document. Those quantities require the agreed Namibian management procedure and management inputs.

Table 1: Comparison of 2025 values from the previous assessment with 2026 values from the updated assessment.

Quan- tity	Previous assessment (2025 value)	Updated assessment (2026 value)	Change (2026 minus 2025)
SSB	490.867	466.951	-23.916
Deple- tion	0.172	0.165	-0.008

Table 1: Comparison of 2025 values from the previous assessment with 2026 values from the updated assessment.

Quantity	Previous assessment (2025 value)	Updated assessment (2026 value)	Change (2026 minus 2025)
B_Bmsy	0.521	0.497	-0.024
Catch	140.001	133.001	-7.000
RY	104.865	98.766	-6.099
Catch_RY	1.335	1.347	0.012

Introduction

This operational update applies the 2025 base-model structure to information available through 2026. Stable model methods are not repeated in full; the emphasis is on new inputs, bridge results, model diagnostics, and sensitivity to the principal structural assumptions considered in the previous assessment.

Fishery and management history

The assessment input contains annual catches from 1964 through 2026. Management-history text and the final 2026 management decision should be inserted following review by MFMR.

`TODO: add the final 2026 TAC decision and any in-season management changes.`

Data

Table 2: Automated validation of the 2026 model inputs.

Check	Result
Annual model years	1964–2026
Weight-at-age dimensions	63×10
Fishery composition dimensions	63×11
CPUE dimensions	63×7
Survey dimensions	63×5
Control EOF sentinel	54321
Data EOF sentinel	12345

Survey data

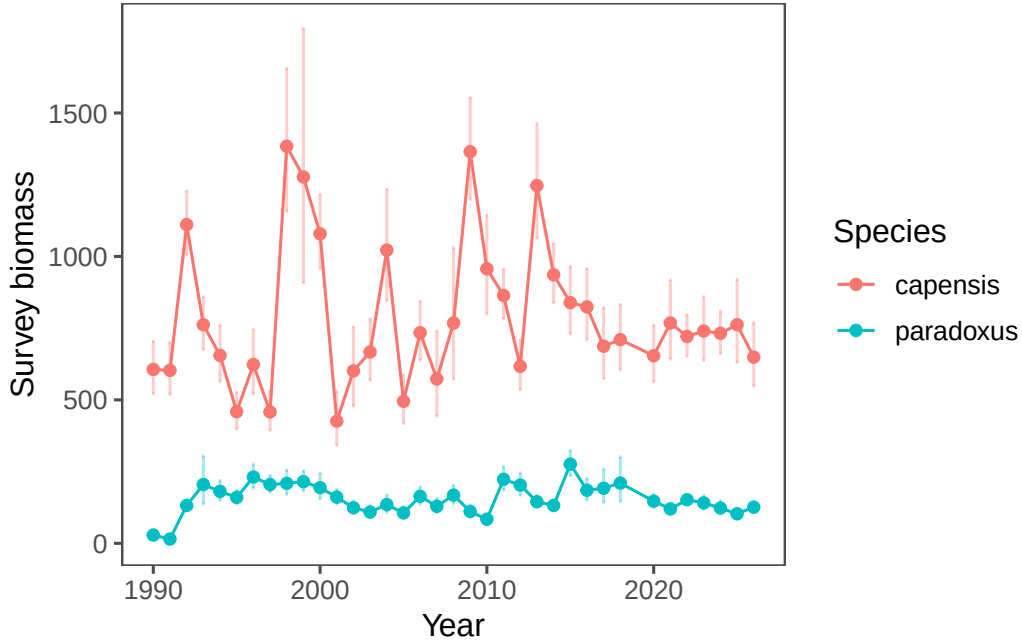


Figure 1: Species-specific survey biomass observations through 2026.

Other data

No additional external data series were introduced in this update. Missing survey observations remain represented as missing or zero according to the existing model input convention; no values were imputed to create artificial annual continuity.

Analytic approach

The age-structured ADMB model and 2025 base assumptions were retained. The control file specifies the terminal year in four locations: the model terminal year, final selectivity period, final recruitment-estimation year, and final Ksp period. All four are set to 2026.

The bridge comparison changes only the data and terminal year. Sensitivities fix survey catchability at 1.0 and 0.5, estimate natural mortality, allow dome-shaped fishery selectivity, and fix stock-recruit steepness at 0.5 and 0.9. Convergence requires a finite objective, maximum gradient no greater than 0.001, and a positive-definite ADMB Hessian.

Results

Bridge comparison

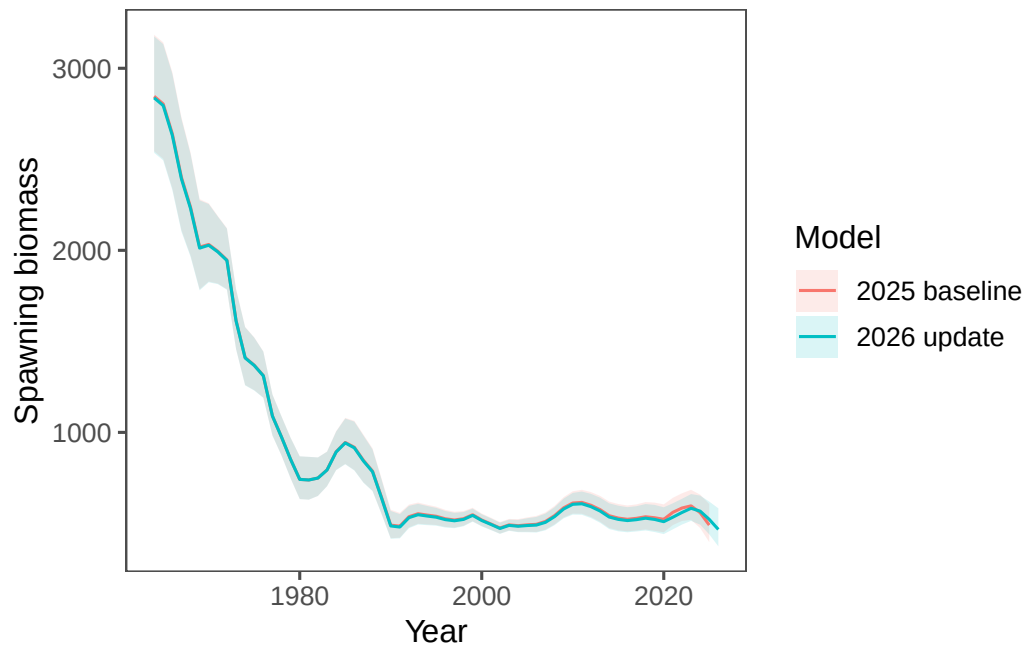


Figure 2: Spawning biomass from the unchanged 2025 model and the 2026 data update.

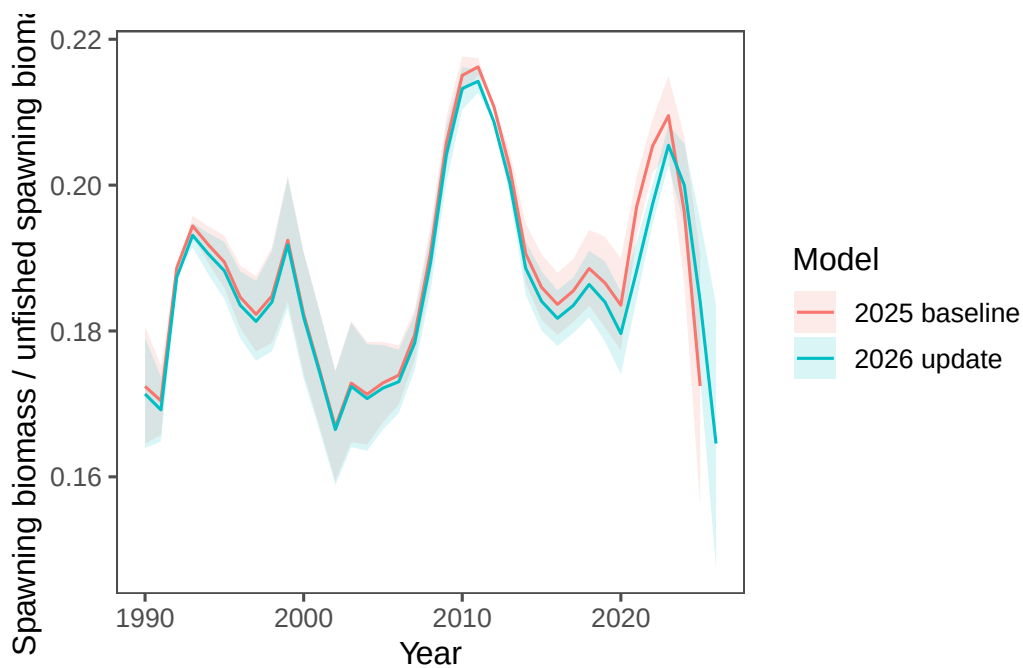
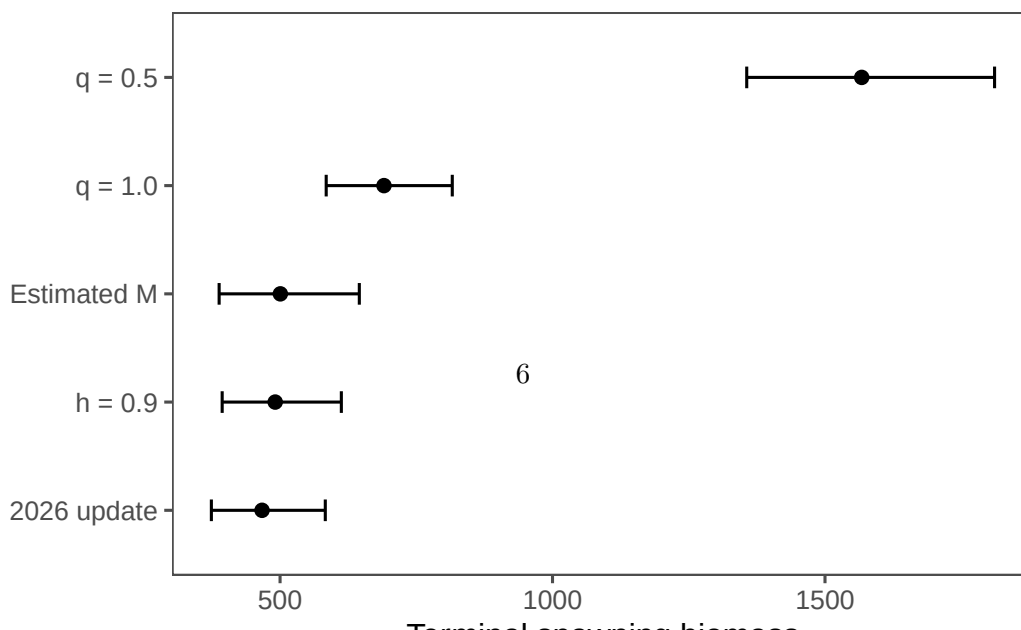


Figure 3: Estimated depletion from the 2025 baseline and 2026 update.

Model diagnostics and sensitivities

Table 3: Convergence diagnostics for the bridge and sensitivity models.

Model	Objective	Maximum gradient	Status
2025 baseline	-44.15933	0.00023	Converged
2026 update	-44.97733	0.00004	Converged
$q = 1.0$	-25.52259	0.00006	Converged
$q = 0.5$	32.89469	0.00003	Converged
Estimated M	-45.62760	0.00007	Converged
Dome selectivity	-46.56746	2.49472	Failed diagnostic criteria
$h = 0.5$	-53.76517	0.00302	Failed diagnostic criteria
$h = 0.9$	-33.47065	0.00008	Converged



Control rule application

Modeling Namibian hake survey data by species

Fisheries stock assessments require data that are reliably collected and compiled. Assessment models should also be configured to match the assumptions associated with the observed data. To account for survey trends between the two hake species, the estimated observation errors were applied in a state-space random-walk model. The observation model applies the observation-error variances $\sigma_{j,t}^2$ for species j in year t . The observed indices $Z_{j,t}$ are fit to latent population trends $\hat{Z}_{j,t}$:

$$\log Z_{j,t} = \log \hat{Z}_{j,t} + \epsilon_{j,t}, \quad \epsilon_{j,t} \sim N(0, \sigma_{j,t}^2).$$

The state equation and associated process-error variance τ_j^2 is:

$$\log \hat{Z}_{j,t+1} = \log \hat{Z}_{j,t} + \eta_{j,t}, \quad \eta_{j,t} \sim N(0, \tau_j^2).$$

The process-error variances may be shared or allowed to vary among indices, and the unobserved species population is estimated as a series of random effects. The model is fit by maximum likelihood in TMB using the [REMA package](#). Species-specific survey observations through 2026 were fit using their supplied CVs as observation-error inputs.

Application to the Namibian hake stocks

The control rule was first applied using survey data and the relative proportions of the two hake species. A more robust formulation responds directly to the *M. paradoxus* biomass signal, independently of *M. capensis*. For this purpose, the survey smoother was fit to *M. paradoxus* alone. The long-term reference level is the geometric mean of predicted biomass from 1990 through 2026.

Figure 5 updates Figure 18 from the 2025 assessment with the 2026 *M. paradoxus* survey observation.

The historical adjustments for three values of the species reactivity parameter γ are shown in Figure 6. Values above the long-term mean are capped at one, so this component only reduces the combined stock advice.

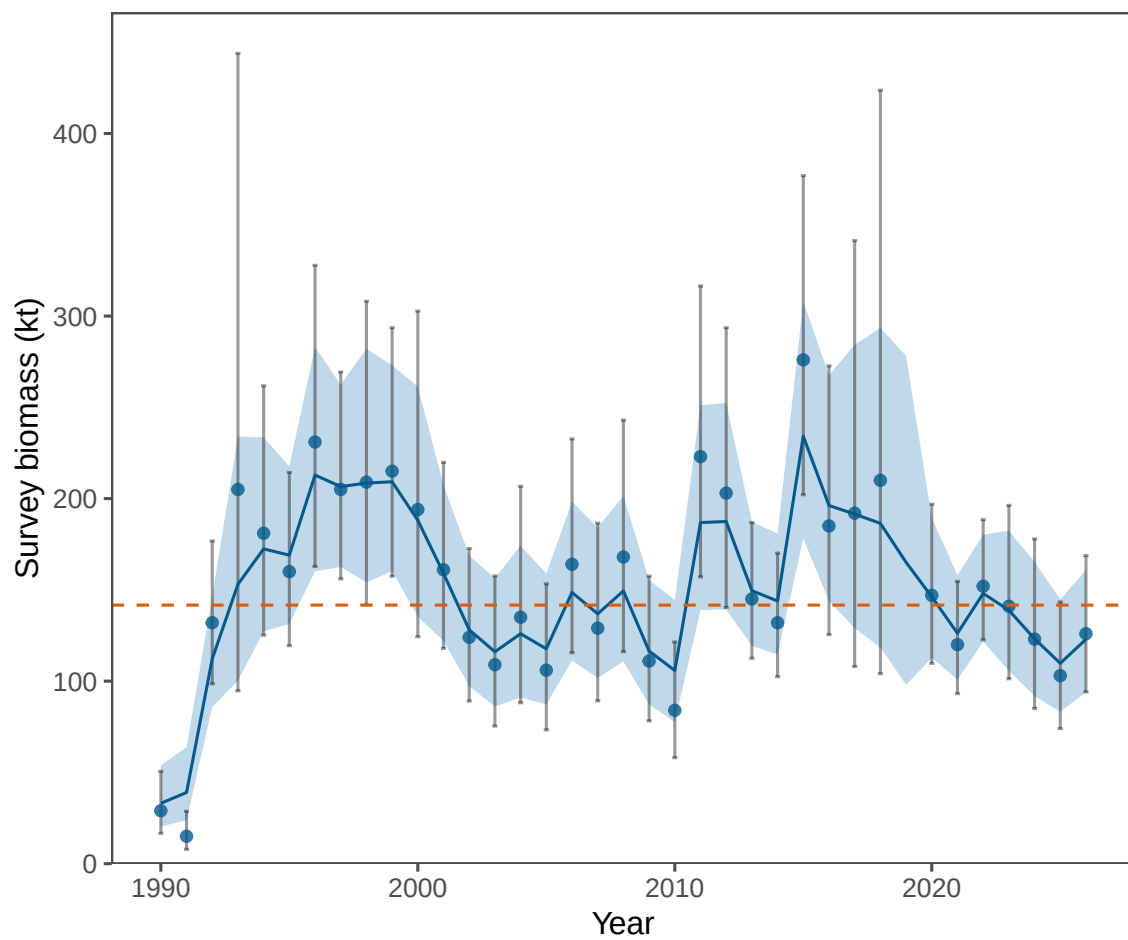


Figure 5: Historical REMA-smoothed biomass estimates for *M. paradoxus* and the geometric mean over 1990–2026.

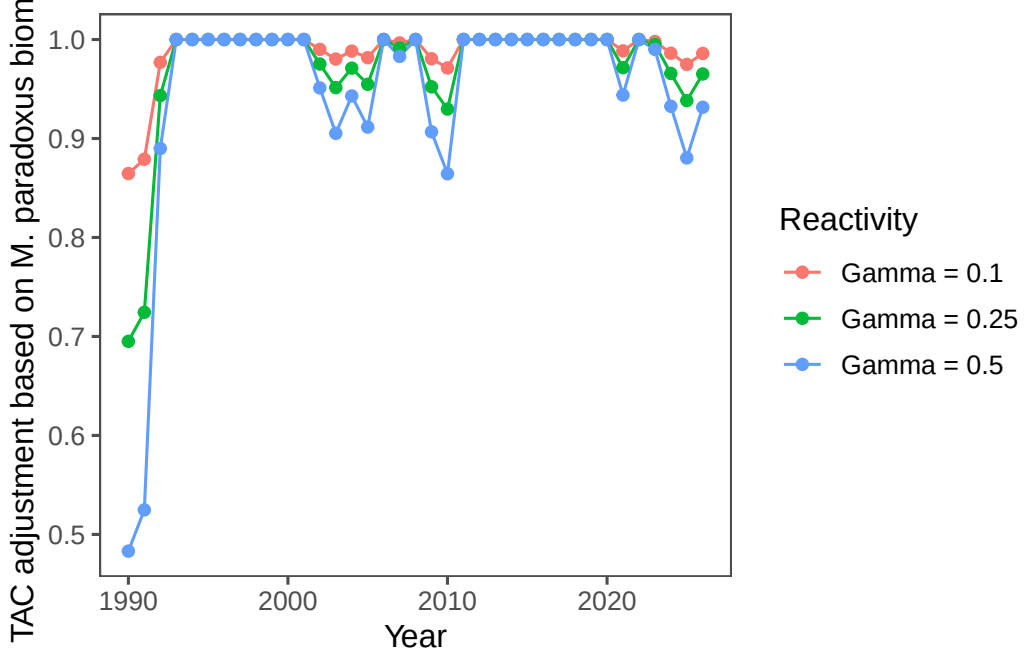


Figure 6: Historical TAC adjustment factors for alternative reactivities to *M. paradoxus* biomass.

Control rule developments

The two-species control rule combines the overall biomass trend from the assessment with the species-specific survey signal. Its purpose is to reduce the exploitation rate when either combined spawning biomass or *M. paradoxus* biomass is low. The specification evaluated in the 2025 report is retained:

$$TAC_y = \begin{cases} 0, & p_y < 0.3\bar{p}, \\ \delta \left(\frac{1}{5} \sum_{i=0}^4 RY_{y-i} \right) f(B_y, \lambda) \min \left(1, \frac{\tilde{p}_y}{\bar{p}} \right)^\gamma, & p_y \geq 0.3\bar{p}, \end{cases}$$

where:

$$f(B_y, \lambda) = \begin{cases} \left(\frac{B_y/B_{MSY}}{0.5} \right)^\lambda, & B_y/B_{MSY} < 0.5, \\ 1, & B_y/B_{MSY} \geq 0.5. \end{cases}$$

The rebuilding factor δ is 0.8 when spawning biomass is below B_{MSY} and 1.0 otherwise. The λ term responds when spawning biomass falls below $0.5B_{MSY}$, while γ controls the response to the *M. paradoxus* biomass ratio.

Table 4 applies the same four λ - γ options shown in the 2025 report to the unchanged 2025 model and the 2026 update.

Table 4: Namibian hake control-rule calculations under the options evaluated in the 2025 report.

Statistic	2025 baseline	2026 update
Recent 5-year average replacement yield	150.754	149.262
Current spawning biomass / BMSY	0.521	0.497
2026 <i>M. paradoxus</i> biomass / long-term mean	0.868	0.868
Option 1: $\lambda = 1.5$, $\gamma = 0.25$	116.404	114.304
Option 2: $\lambda = 2$, $\gamma = 0.25$	116.404	113.990
Option 3: $\lambda = 2$, $\gamma = 0.1$	118.906	116.440
Option 4: $\lambda = 2$, $\gamma = 1$	104.663	102.493

The response curves retained from the 2025 report are shown in Figure 7 and Figure 8. For the 2026 update, *M. paradoxus* predicted biomass is 0.868 of its 1990–2026 geometric mean, and the base-model spawning biomass is 0.497 of B_{MSY} .

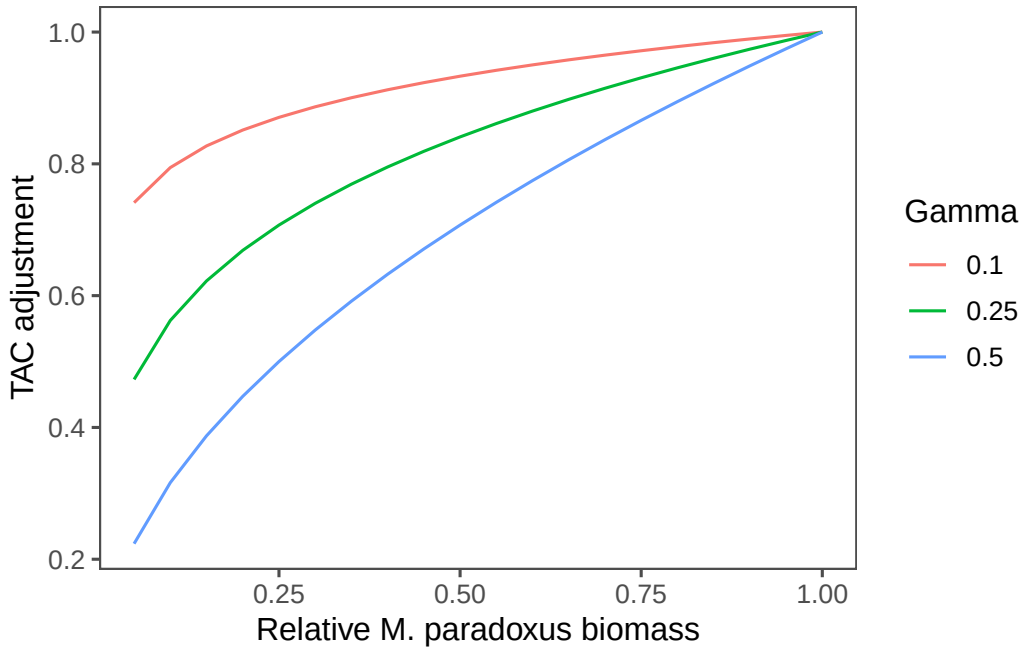


Figure 7: TAC adjustment for alternative values of gamma and relative *M. paradoxus* biomass.

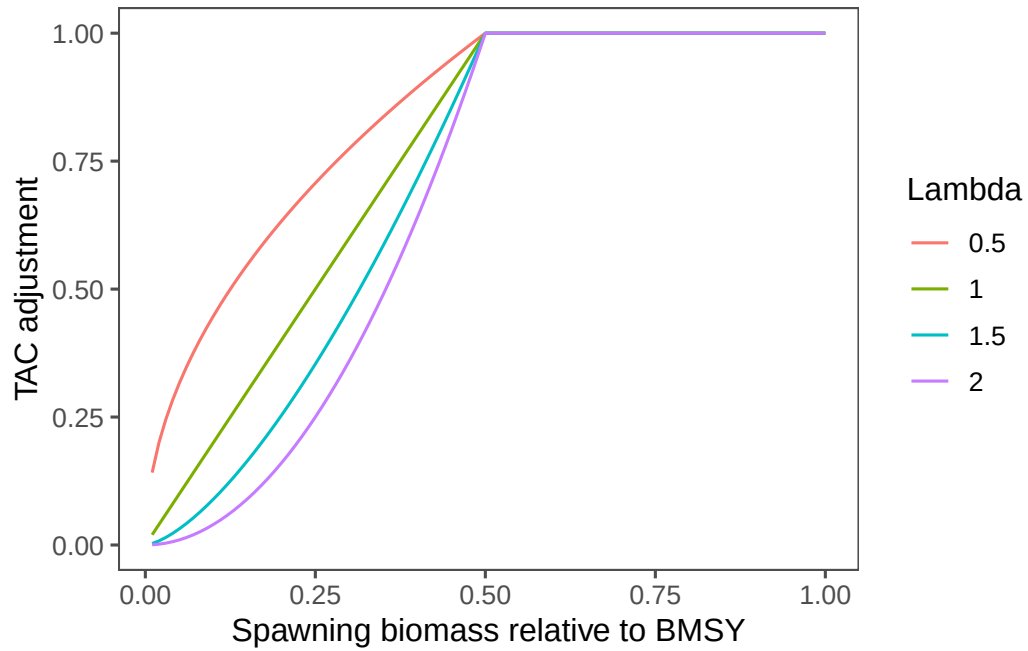


Figure 8: TAC adjustment for alternative values of lambda and spawning biomass relative to BMSY.

Projections

A set of simple 40-year projections was produced from the 2026 base-case model. The scenarios hold catch at 50%, 100%, or 110% of the 2026 catch; increase or decrease catch by 10% for 10 years before returning to the 2026 catch; or set catch to the model-estimated replacement yield. Their effects on catch and depletion are compared in Figure 9.

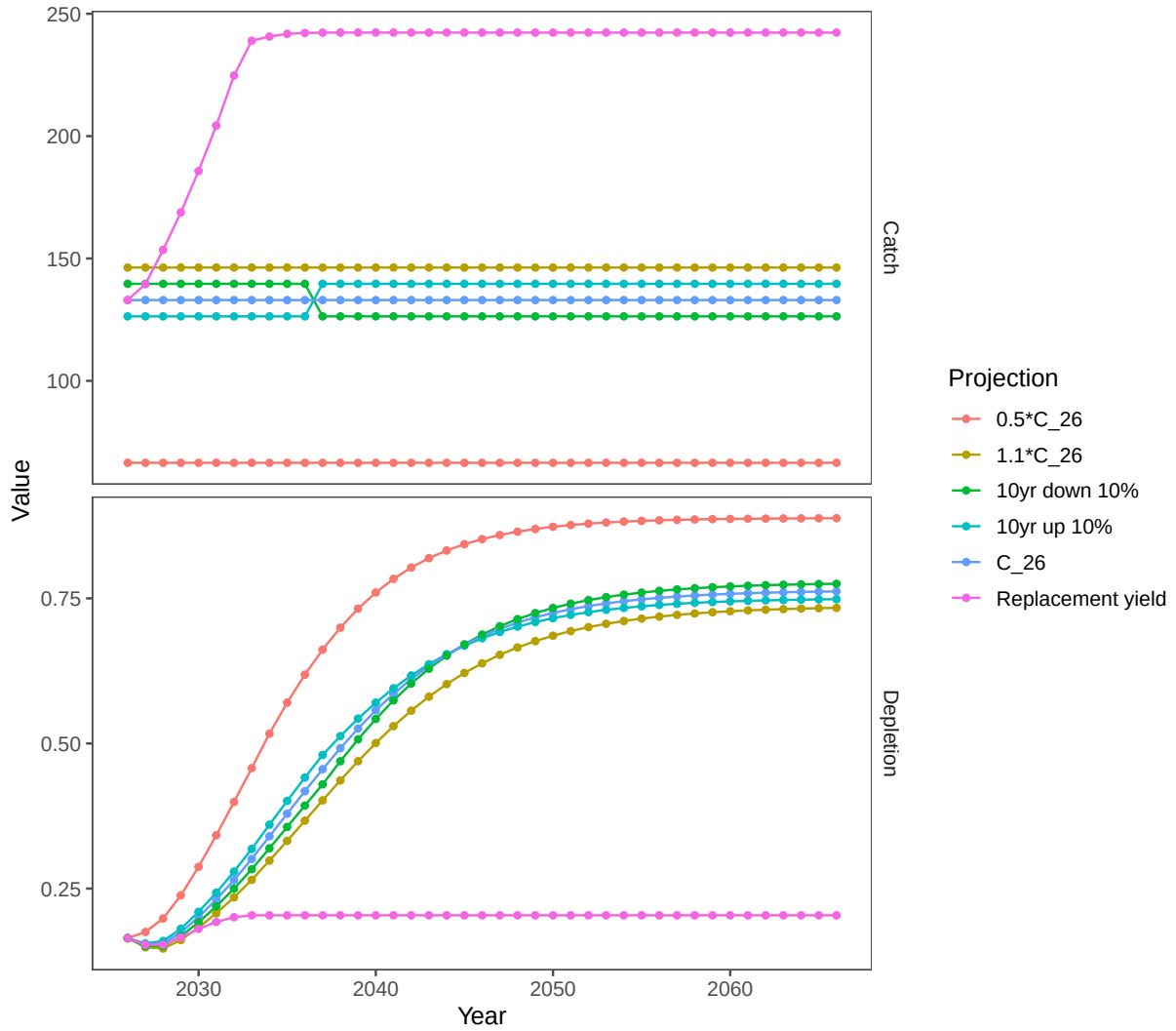


Figure 9: Forty-year projections under catch scenarios based on the 2026 catch. C_{26} is the 2026 catch; the 10-year scenarios alter catch by 10% before returning to C_{26} ; replacement yield sets catch to the model-estimated replacement yield.

Harvest recommendations

The ADMB model produced replacement-yield and long-term reference-point outputs for each converged configuration. A management recommendation is not presented because the final choice among control-rule options and other management inputs was not supplied as part of this update. The calculations in Table 4 reproduce the options evaluated previously and should not be interpreted as adopted TAC advice.

Risk table and recommendation

The 2026 base model passed the declared numerical diagnostics. Structural uncertainty remains material: estimated natural mortality and fixed survey catchability change the inferred scale of the stock, while the dome-shaped selectivity case is not estimable under the current parameterization. The non-converged $h = 0.5$ and dome-selectivity results should not be assigned inferential weight.

Status determination

The base-model estimate of spawning biomass relative to BMSY is reported in Table 1. Formal legal or management status terminology should be supplied by MFMR using the applicable Namibian framework.

Ecosystem considerations

No new ecosystem covariates were incorporated in the assessment likelihood. The species-specific survey series is retained for evaluating the relative signals for *M. paradoxus* and *M. capensis*.

Data gaps and research priorities

1. Resolve the non-identifiability of dome-shaped selectivity before using that alternative for management inference.
2. Profile or otherwise investigate the $h = 0.5$ optimization surface.
3. Document the provenance and preprocessing of each annual input block.
4. Add retrospective analyses and likelihood profiles to the operational workflow.
5. Archive the final management-procedure inputs alongside the accepted assessment.

Acknowledgements

The assessment update was prepared by John Kathena and Jim Ianelli with data provided by the Namibian Ministry of Fisheries and Marine Resources.

Auxiliary files

The assessment source, annual controls, automated validation tests, ADMB template, and model outputs are maintained in the **NamibianHake** repository.